REPORT 4 - NOSE CONE CFD IN ANSYS FLUENT

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Weisshorn – Nose Cone CFD: Workflow and Study A

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1 Summary

Fluent workflow for the Weisshorn nose cone, written as the step-by-step record of study A of 2027_C_FD_REPORT-2_NOSE-CONE-650MM: LV-Haack, 561 mm curved part + 89 mm coupler, $D = 180$ mm, $r_n = 4.5$ mm, zero incidence. Steady 2D axisymmetric RANS (k-ω SST) in Fluent 2024 R2 at $M = 0.95$, 1.2 and 1.47. Sections 3–7: steps; Section 8: results; Section 9: comparison with Python. Study B will be added after its runs.

Table 1: Study A: drag coefficient of the nose (`wall_nose`, tip to shoulder), $A_{\text{ref}} = \pi D^2/4 = 0.025\,45\ \text{m}^2$.

	$M = 0.95$	$M = 1.2$	$M = 1.47$
Pressure C_D	0.0042	0.0948	0.0963
Viscous C_D	0.0248	0.0224	0.0207
Total C_D	**0.0291**	**0.1172**	**0.1170**
Drag force	42.3 N	254.8 N	349.4 N
Solver	pressure-based	density-based	density-based

- **Supersonic**: $C_D \approx 0.117$, four fifths of it wave drag. Density-based with Roe-FDS, about 3000 iterations.
- **Transonic**: pressure drag ≈ 0, friction 85 % of the total. Only the pressure-based Coupled solver converged.
- **Against Python**: wave drag $+2.4\,\%$, friction within 5 %. The transonic ramp gives 0.077 where Fluent gives 0.004, so the report-2 drag is conservative below $M = 1$.

2 Case definition

2.1 The noses

Table 2: The two noses of report 2. Lengths in mm; x is measured from the rounded tip.

	Study A (this report)	Study B (to be run)
Profile	LV-Haack	Von Kármán
Nominal curved length L, fineness L/D	561, 3.117	630, 3.5
Coupler length c, total length	89, 650	20, 650
Tip radius r_n	4.5	4.5
End of the curved part (shoulder), x	558.65	625.42
End of the coupler, x	647.65	645.42

Both noses are solids of revolution in mm, with x along the axis and the tip at $x = 0$. The 4.5 mm tip rounding shortens the curved part by 2.35 mm (A) and 4.58 mm (B). STEP files: `CFD/geometry_650/nose_lvhaack_L561_c89_rn4.5mm.step` (A), `nose_vonkarman_L630_c20_rn4.5mm.step` (B).

2.2 Flight conditions

Table 3: Flight conditions of the three runs (altitude above sea level).

M	altitude	p_∞	T_∞	type of flow
0.95	928 m	90 666 Pa	282.1 K	transonic: just below the speed of sound
1.20	1478 m	84 789 Pa	278.5 K	supersonic, close to $M = 1$
1.47	2189 m	77 649 Pa	273.9 K	supersonic, top speed of the flight

The inputs are M, p_∞ and T_∞, from the report-1 trajectory. On the report-2 trajectory of nose A the same Mach numbers are reached within 14 m of altitude and 0.2 % in pressure; its top speed is $M = 1.457$ (2182 m), so the $M = 1.47$ run is the top-speed point ($+1\,\%$).

Why the speed matters. Pressure waves travel upstream at the speed $a - V$ relative to the rocket:

- $M < 1$: $a - V > 0$, the air ahead feels the nose coming. No bow shock; a small supersonic zone forms at the shoulder and ends in a weak shock (transonic flow).
- $M > 1$: $a - V < 0$, the air ahead is undisturbed until a bow shock just in front of the tip. The high pressure behind it, acting on the front of the nose, is the wave drag.

What the runs give. The drag on `wall_nose` (tip to shoulder), split by Fluent into pressure and viscous parts:

$$C_D = \frac{F_x}{q_\infty A_{\text{ref}}}, \qquad q_\infty = \tfrac{1}{2}\rho_\infty V_\infty^2, \qquad A_{\text{ref}} = \frac{\pi D^2}{4}. \tag{1}$$

2.3 Symbols

a speed of sound; A_{ref} reference area $\pi D^2/4$; C_D drag coefficient; C_f friction coefficient; C_p pressure coefficient; CFL Courant number; c_p specific heat at constant pressure; D body diameter (180 mm); F_x force along the rocket axis; h_1 height of the first cell at the wall; I turbulence intensity; k turbulent kinetic energy; M Mach number; p pressure; T_0 stagnation (total) temperature; $q = \frac{1}{2}\rho V^2$ dynamic pressure; R gas constant of air; Re Reynolds number; r_n tip radius; T temperature; u_τ friction velocity; V flow speed; y^+ wall distance in viscous units; γ ratio of specific heats; δ boundary-layer thickness; μ viscosity of the air; μ_t turbulent (eddy) viscosity; ρ density; τ_w wall shear stress; ω turbulence frequency (specific dissipation rate). Subscript ∞: the undisturbed air far from the rocket (free stream).

3 Step 1: Geometry

The flow domain is a flat rectangle in the xy plane, above the rocket axis ($y \geq 0$), with the outline of the nose cut out of its lower edge (Figure 1). It is built in Discovery, inside Workbench, from the design STEP file.

3.1 Steps

1. **Workbench.** Add a **Fluid Flow (Fluent)** system. Before importing anything, click the **Geometry** cell and set **Properties** → **Analysis Type** = **2D**. Without this setting, Fluent does not offer the axisymmetric option used below.
2. **Import.** **Geometry** → **Import Geometry** → **Browse**, choose `CFD/geometry_650/nose_lvhaack_L561_c89_rn4.5mm.step` and open it in **Discovery** (Figure 2a). Use the STEP file, not the STL: the STL is made of small flat triangles, and cutting, naming edges and meshing do not work cleanly on it.
3. **Extend the body.** With **Pull**, drag the flat end face of the coupler by 1800 mm, from $x = 647.65$ mm to $x = 2447.65$ mm. The body is now a long cylinder that reaches the end of the domain.
4. **Draw the domain.** In the XY plane, sketch the rectangle from $x = -3600$, $y = 0$ to $x = 2447.65$, $y = 3600$ (mm) and turn it into a surface.

5. **Cut out the body. Combine → Subtract from target**: the rectangle is the target, the solid is the tool that cuts. Delete the solid and any piece left inside the body. One surface remains, with the half-outline of the nose along its lower edge (Figure 2b). The nose and cylinder edges are already separate, because they are separate faces in the STEP file.
6. **Save the project** (**File → Save As**). Until it is saved, Workbench keeps the geometry in a temporary folder and loses it if it restarts. The study-A domain is stored as `CFD/geometry_650/studyA_domain2D_2024R2.dsco`.
7. **Name the edges.** In Meshing, select each edge, right-click → **Create Named Selection**, and use exactly the names of Table 4. Fluent turns them into boundaries with the same names, and the boundary conditions of Step 3 are set on them (Figure 3). A typing error (`wal_nose`) means renaming the zone in Fluent later.

Table 4: Named edges of the study-A domain (Figure 1).

Name	Edge	Position [m]
`inlet`	left (upstream) edge	$x = -3.6$, y from 0 to 3.6
`farfield`	top edge	$y = 3.6$, x from -3.6 to 2.44765
`outlet`	right (downstream) edge	$x = 2.44765$, y from 0.09 to 3.6
`axis`	rocket axis in front of the tip	$y = 0$, x from -3.6 to 0
`wall_nose`	curved nose, rounded tip included	x from 0 to 0.55865
`wall_body`	cylinder: coupler and its extension	$y = 0.09$, x from 0.55865 to 2.44765

Check. The surface lies entirely above the axis ($y \geq 0$), the x axis is the rocket axis, the tip is at the origin, and every edge has exactly one name. When the mesh is read into Fluent, the console lists one fluid zone and these six boundaries.

3.2 Why the domain looks like this

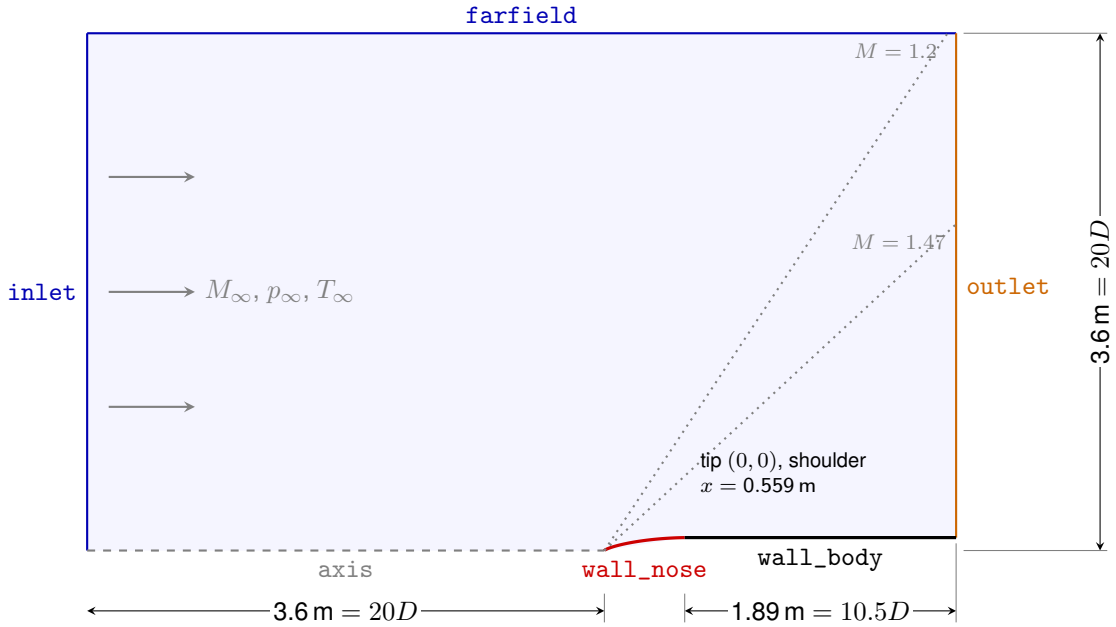
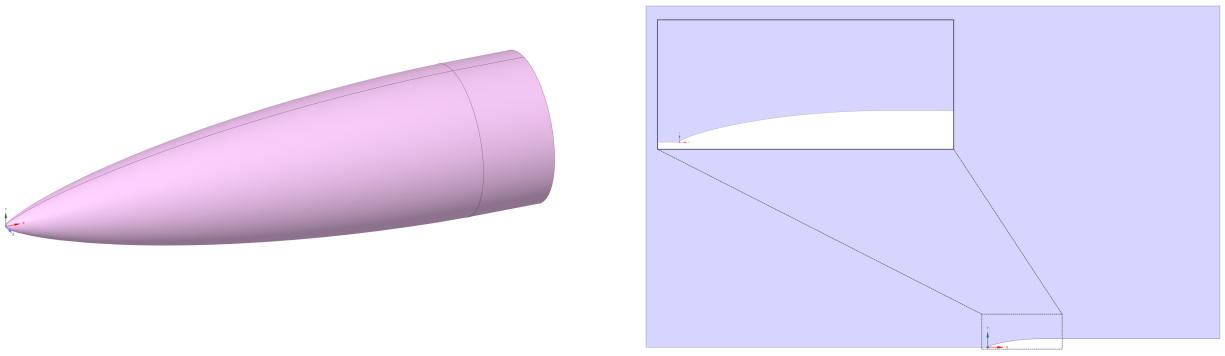


Figure 1: Study-A flow domain to scale, with the named edges. The nose is the small bump on the lower edge. Dotted: the direction in which the bow shock ends up far from the body, at the angle $\arcsin(1/M)$.

- **Only a 2D half-plane (axisymmetric).** With no angle of attack the flow is the same in every plane through the rocket axis, so the half-plane $y \geq 0$ is enough. Fluent counts each wall element as a ring, $\mathrm{d}A = 2\pi y \, \mathrm{d}s$, and reports the forces of the full body.
- **Far boundaries 3.6 m ($20D$) ahead and above.** The undisturbed air is imposed there, so the nose must have no effect at that distance. Above $M = 1$ the air ahead of the bow shock is undisturbed anyway; below $M = 1$ the disturbance fades with distance, and $20D$ is the usual margin.
- **Outlet 1.89 m ($10.5D$) behind the shoulder.** Far enough for the flow along the cylinder to settle before it leaves. Far from the body the bow shock lies at $\mu = \arcsin(1/M)$: 56.4° at $M = 1.2$, 42.9° at $M = 1.47$ (dotted lines). At $M = 1.47$ it leaves through the outlet; at $M = 1.2$ it reaches the top edge at $x = 3.6\,\mathrm{m} / \tan\mu \approx 2.4\,\mathrm{m}$, where the far-field condition lets it out without reflection (Section 5.3).
- **The cylinder runs out of the domain.** A body ending inside the domain would have a flat base with an unsteady wake, which a steady run cannot settle and which, below $M = 1$, also affects the nose. Running the cylinder out gives the drag of the nose alone; its length is set by the domain, not by the rocket.



(a) Starting geometry: the design solid from the STEP file.

(b) Final geometry: the 2D flow domain after the cut, with the nose region enlarged.

Figure 2: Geometry in Discovery (Ansys 2024 R2).

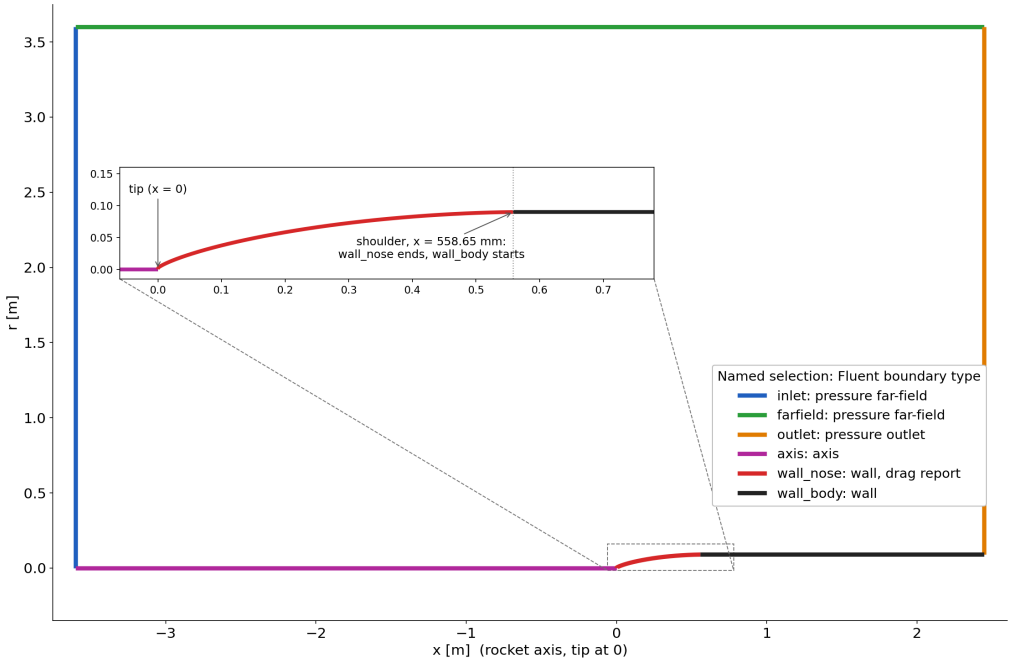


Figure 3: The six named edges and the boundary type each one gets in Fluent, drawn from the boundary zones of the Fluent mesh. Insert: the nose, where `wall_nose` ends and `wall_body` starts.

4 Step 2: Mesh

The mesh is made in Ansys Meshing 2024 R2. Close to the two walls it has thin quadrilateral inflation layers to capture the boundary layer; the rest of the domain is filled with triangles, made smaller along the nose and at the tip (Figure 4).

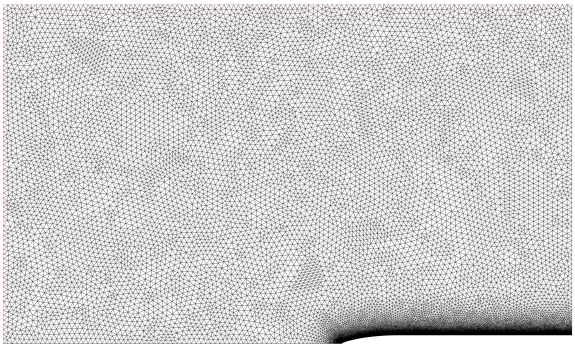
4.1 Steps

1. Open the **Mesh** cell. In **Mesh** → **Details**: **Physics Preference** = CFD, **Solver Preference** = Fluent, **Element Order** = Linear, **Element Size** = 0.05 m, **Growth Rate** = 1.1, **Smoothing** = High.
2. **Insert** → **Method** on the surface: triangles. A quadrilateral-dominant fill failed on this surface.
3. **Insert** → **Sizing** on edges: `wall_nose` 1 mm; `wall_body` 1 mm with **Bias Type** towards the shoulder and **Bias Factor** = 3, so the cells are small at the shoulder and grow towards the outlet.
4. **Insert** → **Sizing** on the surface, **Type** = **Sphere of Influence**: centre at the origin (the tip), radius 15 mm, element size 0.2 mm.
5. **Insert** → **Inflation** on the surface, boundary = `wall_nose` and `wall_body`: **Inflation Option** = **First Layer Thickness**, first layer 1 µm, 45 layers, growth rate 1.2, **Collision Avoidance** = **Layer Compression**.
6. **Generate**, then look at the quality: **Mesh** → **Quality** → **Mesh Metric** = **Skewness**, then **Orthogonal Quality** (Section 4.3). Close Meshing and **Update** the Mesh cell so that Fluent receives the mesh with the named selections.

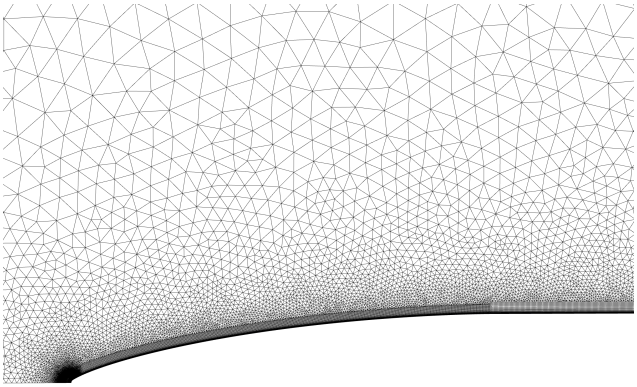
Check. Maximum skewness below 0.95 and minimum orthogonal quality well above 0.01; in Fluent, **Domain** → **Mesh** → **Check** passes with no negative volumes. The study-A mesh has 169 674 cells (Table 5).

Table 5: Study-A mesh.

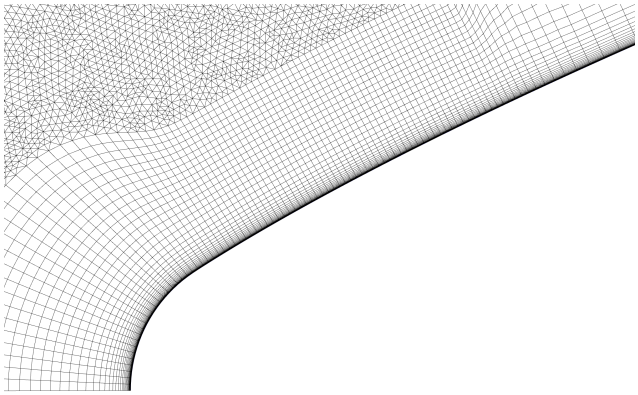
Cells	169 674
Faces / nodes	312 923 / 143 250
Faces on `wall_nose` / `wall_body`	639 / 1889
Faces on `axis` / `inlet` / `farfield` / `outlet`	205 / 72 / 121 / 138
Maximum skewness (Meshing)	0.92
Minimum orthogonal quality (Meshing / Fluent)	0.237 / 0.077
Maximum aspect ratio	1660 (first inflation layer at the shoulder)
Fluent mesh check	passed, no negative volumes



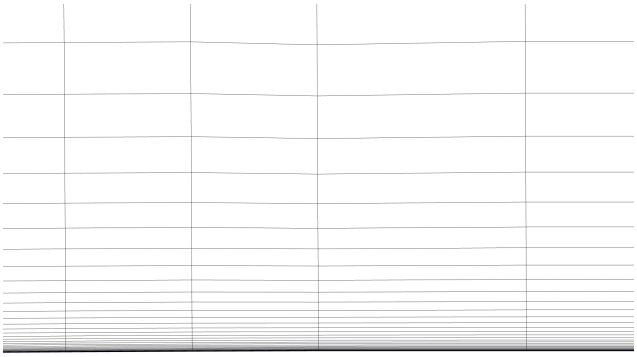
(a) Whole domain.



(b) Nose and shoulder.



(c) Tip, about 20 mm window: inflation layers around the rounded tip.



(d) Inflation layers at the shoulder, about 5 mm window.

Figure 4: Study-A mesh, as read by Fluent.

4.2 Why these sizes

First cell height. SST is solved down to the wall, without wall functions, so the first cell must sit in the viscous layer: $y^+ = \rho u_\tau y/\mu \approx 1$. Flat-plate estimate at the shoulder ($x = 0.559$ m), with the free stream of Section 5.2:

$$\mathrm{Re}_x = \frac{\rho V x}{\mu}, \qquad C_f \approx 0.026\,\mathrm{Re}_x^{-1/7}, \qquad u_\tau = V\sqrt{\frac{C_f}{2}}, \qquad h_1 = \left.\frac{y^+\,\mu}{\rho\,u_\tau}\right|_{y^+=1}. \tag{2}$$

This gives $h_1 = 1.38$, 1.16 and 1.02 µm at $M = 0.95$, 1.2 and 1.47, so one mesh with $h_1 = 1$ µm covers all three. Fluent's y^+ after the runs stays below 0.71 on the nose and 0.30 on the cylinder (Figure 5).

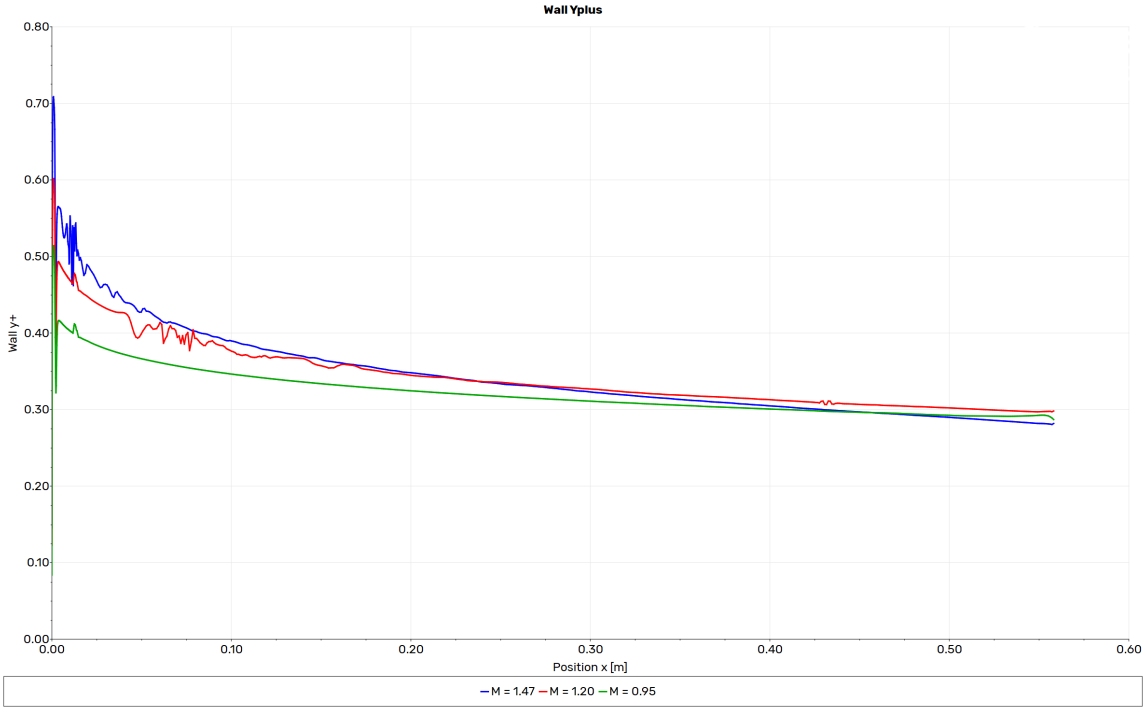


Figure 5: Wall y^+ along `wall_nose` at the three Mach numbers (Fluent XY plot). It stays below 1 everywhere, as Equation (2) intended.

Inflation thickness.

$$H = h_1 \frac{1.2^{45} - 1}{1.2 - 1} = 18.3\,\text{mm}, \qquad h_{45} = h_1 \times 1.2^{44} = 3.05\,\text{mm}. \tag{3}$$

The boundary layer at the shoulder is $\delta \approx 0.37\,x\,\mathrm{Re}_x^{-1/5} = 7.5$–$8.0$ mm, so it lies entirely inside the inflation layers.

Tip. The tip arc $\pi r_n/2 = 7.07$ mm gets about 35 cells of 0.2 mm. This is where the bow shock forms and the wall pressure changes fastest (Figure 11).

4.3 Mesh quality

Table 6: Mesh quality of study A (**Mesh** → **Quality** → **Mesh Metric** in Meshing, mesh check in Fluent).

Metric	Meaning	Limit	Study A
Skewness	how far the cell angles are from the ideal shape (0 = perfect, 1 = flattened cell)	maximum < 0.95	0.92, acceptable
Orthogonal quality	how well each face lines up with the line between the two cell centres (1 = perfect, 0 = worst)	minimum > 0.01 (Fluent)	0.237 (Meshing), 0.077 (Fluent)
Aspect ratio	cell length / cell height	none for inflation layers	1660, first layer at the shoulder

Table 7: How Ansys Meshing rates skewness and orthogonal quality.

Skewness	0–0.25 excellent	0.25–0.50 very good	0.50–0.80 good	0.80–0.94 acceptable	0.95–0.97 bad	0.98–1 unacceptable
Orthogonal quality	0–0.001 unacceptable	0.001–0.14 bad	0.15–0.20 acceptable	0.20–0.69 good	0.70–0.95 very good	0.95–1 excellent

- The worst cells are near the outlet, where the last inflation layers on `wall_body` meet the triangles, 1.4 m behind the nose; around the nose the cells are close to ideal (Figure 4c).
- A bias factor of 10 on `wall_body` stretched those cells to a skewness of 0.985 (38 cells, “unacceptable”); a bias factor of 3 brought the maximum down to 0.92.
- Long, thin first layers (aspect ratio about 1000) are normal for $y^+ \approx 1$: across the boundary layer the velocity changes about a thousand times faster than along the wall.

5 Step 3: Models and boundary conditions

5.1 Models

Steps. Launch Fluent from the Setup cell: **Dimension** 2D, **Double Precision**, 6 processes. Then:

1. **General**: **Type** = **Density-Based** ($M = 1.2$, 1.47) or **Pressure-Based** ($M = 0.95$, see Section 6); **Time** = **Steady**; **2D Space** = **Axisymmetric**.
2. **Models** → **Energy**: on. **Models** → **Viscous**: **k-omega**, **SST**, default constants, compressibility effects off. With the pressure-based solver, also tick **Viscous Heating** in the same panel.
3. **Materials** → **air**: **Density** = **ideal-gas**, **Viscosity** = **sutherland** (three-coefficient method). Keep Fluent’s $c_p = 1006.43\,\mathrm{J/(kg\,K)}$, thermal conductivity 0.0242 W/(m K) and molecular weight 28.966.
4. **Operating Conditions**: **Operating Pressure** = 0 Pa. Fluent works with gauge pressure, $p - p_{\mathrm{op}}$; with $p_{\mathrm{op}} = 0$ every pressure you type or read is absolute, which is the simplest choice when the pressure changes by tens of per cent across the domain.

What Fluent solves. Steady, time-averaged (RANS) conservation of mass, momentum and energy for a compressible gas, closed by the ideal-gas law and Sutherland’s law (Fluent’s air: $R = 287.04\,\mathrm{J/(kg\,K)}$, $\gamma = 1.399$):

$$p = \rho R T, \qquad \mu(T) = \mu_{\mathrm{ref}} \left(\frac{T}{T_{\mathrm{ref}}}\right)^{3/2} \frac{T_{\mathrm{ref}} + S}{T + S}, \tag{4}$$
$$\mu_{\mathrm{ref}} = 1.716 \times 10^{-5}\,\mathrm{Pa\,s}, \quad T_{\mathrm{ref}} = 273.11\,\mathrm{K}, \quad S = 110.56\,\mathrm{K}.$$

k-ω SST adds two equations, for k and ω, which give the eddy viscosity $\mu_t \approx \rho k/\omega$. It behaves like k-$\omega$ near the wall (solved down to $y^+ \approx 1$) and like k-ϵ further out, where k-ω is too sensitive to the free-stream values.

5.2 Free stream

Everything follows from M, p_∞ and T_∞ (Table 3):

$$\rho_\infty = \frac{p_\infty}{R T_\infty}, \quad V_\infty = M\sqrt{\gamma R T_\infty}, \quad q_\infty = \tfrac{1}{2}\rho_\infty V_\infty^2, \quad \mathrm{Re}_x = \frac{\rho_\infty V_\infty x}{\mu_\infty}, \quad T_0 = T_\infty\left(1 + \frac{\gamma - 1}{2} M^2\right). \tag{5}$$

Only M, p_∞ and T_∞ are typed into Fluent. `CFD/studyA/freestream_bc.py` recomputes Table 8, and Fluent's reference values agree to five or six digits.

Table 8: Free stream of the three runs.

	$M = 0.95$	$M = 1.2$	$M = 1.47$
Static pressure p_∞ (input)	90 666 Pa	84 789 Pa	77 649 Pa
Static temperature T_∞ (input)	282.1 K	278.5 K	273.9 K
Density ρ_∞	1.1197 kg/m^3	1.0606 kg/m^3	0.9876 kg/m^3
Velocity V_∞	319.7 m/s	401.3 m/s	487.5 m/s
Viscosity μ_∞	1.760×10^{-5} Pa s	1.743×10^{-5} Pa s	1.720×10^{-5} Pa s
Dynamic pressure q_∞	57 238 Pa	85 407 Pa	117 371 Pa
Re_x at the shoulder ($x = 0.559$ m)	$1.14 \cdot 10^7$	$1.37 \cdot 10^7$	$1.56 \cdot 10^7$
Stagnation temperature T_0	332.9 K	358.5 K	392.0 K

5.3 Boundary conditions

Steps (**Setup** → **Boundary Conditions**; Table 9):

1. `inlet`: **Type** = **pressure-far-field**. **Gauge Pressure** $= p_\infty$, **Mach Number** $= M$, flow direction $(1, 0)$; **Thermal** tab: **Temperature** $= T_\infty$; turbulence as below. Then **Copy** these settings to `farfield`.
2. `outlet`: **pressure-outlet**. **Gauge Pressure** $= p_\infty$, **Backflow Total Temperature** $= T_\infty$ (see the note below), backflow direction normal to the boundary, **Average Pressure Specification** and **Target Mass Flow Rate** off; turbulence as below.
3. `axis`: **axis**. `wall_nose` and `wall_body`: **wall**, stationary, no slip, **Thermal** → **Heat Flux** = 0, roughness height 0.

Table 9: Boundary conditions.

Zone	Fluent type	Values
`inlet`, `farfield`	pressure far-field	p_∞, M, T_∞, direction $(1, 0)$; $I = 1\,\%$, $\mu_t/\mu = 10$
`outlet`	pressure outlet	$p = p_\infty$; backflow: total temperature T_∞, normal direction, $I = 1\,\%$, $\mu_t/\mu = 10$
`axis`	axis	none
`wall_nose`, `wall_body`	wall	no slip ($\mathbf{u} = 0$), adiabatic ($q_w = 0$), smooth

Pressure far-field. It imposes the free stream, but only the part that physically comes from outside. At each face Fluent takes one wave quantity from outside and one from the cell inside (u_n the velocity normal to the boundary, positive outwards):

$$R_\infty = u_{n,\infty} - \frac{2a_\infty}{\gamma - 1} \ \text{(outside)}, \qquad R_i = u_{n,i} + \frac{2a_i}{\gamma - 1} \ \text{(inside)}, \tag{6}$$

$$u_n = \tfrac{1}{2}(R_i + R_\infty), \qquad a = \tfrac{\gamma - 1}{4}(R_i - R_\infty). \tag{7}$$

Flow direction and temperature come from outside where air enters and from inside where it leaves, so waves from the body leave without reflecting. Where the inflow is supersonic (`inlet` at $M = 1.2$ and 1.47) the free stream is imposed completely.

Pressure outlet. Subsonic outflow: $p = p_\infty$ is imposed and the rest is taken from inside. Supersonic outflow: everything is taken from inside and the set pressure is ignored. At $M = 1.2$ and 1.47 the outflow is supersonic except in the boundary layer; at $M = 0.95$ it is subsonic over the whole outlet, where a fixed pressure reflects waves (Section 6). Backflow values only act where air re-enters.

Axis and walls. Axis: $v = 0$ and nothing changes across it. Walls: no slip, $\mathbf{u} = 0$; adiabatic, $q_w = -k\,\partial T/\partial n = 0$, so the wall warms to about T_0, which sets the near-wall density and the friction; smooth.

Turbulence at the inlet, far field and outlet. **Intensity and Viscosity Ratio**, with $I = 1\,\%$ and $\mu_t/\mu = 10$ (calm air). Fluent turns them into

$$k = \tfrac{3}{2}\,(I V_\infty)^2, \qquad \omega = \frac{\rho k}{\mu}\left(\frac{\mu_t}{\mu}\right)^{-1}. \tag{8}$$

Fluent's defaults for a new boundary, $k = 1\,\mathrm{m^2/s^2}$ and $\omega = 1\,\mathrm{s^{-1}}$, give $\mu_t = \rho k/\omega \approx 1\,\mathrm{Pa\,s} \approx 60\,000\,\mu$, which would swamp the boundary layer: set the values in the panel.

Backflow temperature. Set to T_∞; T_0 would be the consistent value. It only acts on air re-entering through the outlet and is listed as an open item (Section 10).

5.4 Reference values and drag coefficient

Steps. **Reference Values** → **Compute from** = `inlet`. This sets the reference density, velocity, pressure and temperature to the free stream of the run. Then **Area** = 0.025 45 m² and **Length** = 0.18 m (used for moments only). Redo **Compute from** after every change of the inlet values. Then **Report Definitions** → **New** → **Force Report** → **Drag**: zone `wall_nose`, direction $(1, 0)$, name `cd-nose`, printed, plotted and written to a file at every iteration.

Definition.

$$F_x = \underbrace{\int_{\text{wall_nose}} (p - p_{\text{ref}})\,\mathbf{n}\cdot\mathbf{e}_x\,\mathrm{d}A}_{\text{pressure part}} + \underbrace{\int_{\text{wall_nose}} \boldsymbol{\tau}_w\cdot\mathbf{e}_x\,\mathrm{d}A}_{\text{viscous part}}, \qquad \mathrm{d}A = 2\pi y\,\mathrm{d}s, \tag{9}$$

$$C_D = \frac{F_x}{q_\infty A_{\text{ref}}}, \qquad A_{\text{ref}} = \frac{\pi D^2}{4} = 0.025\,45\,\mathrm{m^2}. \tag{10}$$

Fluent reports both parts. With $p_{\text{ref}} = p_\infty$ the pressure part is the wave drag; with $p_{\text{ref}} = 0$ it would also contain $p_\infty \pi R^2 \approx 2\,\mathrm{kN}$.

6 Step 4: Solver settings for each speed

Study A uses the density-based solver for the two supersonic runs and the pressure-based solver at $M = 0.95$, where the density-based solver did not converge.

6.1 Density-based solver: supersonic runs

How it works. ρ, $\rho\mathbf{u}$ and ρE of every cell are updated together, marching in pseudo-time until nothing changes; k and ω are solved after them in each iteration.

Roe-FDS flux. It splits the jump between two cells into pressure waves and flow, and takes each from its upstream side: shocks come out sharp (2–3 cells) and without wiggles.

Courant number.

$$\Delta t = \mathrm{CFL}\,\frac{\Delta x}{|u| + a}. \tag{11}$$

CFL 1 while the shock forms, then 5; lower it to 3 if the residuals grow.

Order. First-order upwind, $\phi_f = \phi_c$, is robust but smears the shock; second-order upwind, $\phi_f = \phi_c + \nabla\phi \cdot \mathbf{r}$, uses least-squares cell-based gradients. Only second-order results are used.

Steps. **General**: **Density-Based**. **Methods**: **Formulation** = Implicit, **Flux Type** = Roe-FDS, **Gradient** = Least Squares Cell Based, **Flow**, **Turbulent Kinetic Energy** and **Specific Dissipation Rate** = First Order Upwind (Second Order Upwind after the switch). **Controls**: **Courant Number** = 1, then 5.

Why. Shocks dominate and information only travels downstream, which is what the density-based solver is built for. It converged in 3000–3200 iterations (Section 7).

6.2 Pressure-based solver: transonic run

How it works. The unknowns are velocity and pressure; the pressure equation comes from the mass equation, and $\rho = p/(RT)$. **Coupled** solves momentum and pressure together; the **Pseudo Time Method** damps the changes between iterations, with an automatic step.

Why at $M = 0.95$. The flow is subsonic everywhere, waves travel upstream too, and the weak shock at the shoulder moves with the pressure behind it. The density-based C_D never settled:

- second order, CFL 3: 0.009–0.053;
- non-reflecting outlet: 0.030–0.040;
- CFL 1: 0.016–0.031.

The pressure-based solver converged in 1200 iterations, with residuals still falling (10^{-6} to 10^{-12}).

Viscous Heating on. Friction heats the boundary layer through the energy term

$$\Phi = \boldsymbol{\tau} : \nabla \mathbf{u}. \tag{12}$$

The density-based solver always includes it; in the pressure-based solver it has to be ticked. Without it the adiabatic wall stays near T_∞ instead of $\approx T_0$, and the friction is wrong.

Residual criteria off. Pressure-based residuals are scaled by their first iterations: they fell below 10^{-3} at iteration 7 and Fluent stopped the run. Set all criteria to 10^{-12}.

Steps. **General**: **Pressure-Based**. **Models** → **Viscous** → **Options**: **Viscous Heating**. **Methods**: **Scheme** = Coupled, **Pseudo Time Method** on (automatic time step), **Gradient** = Least Squares Cell Based, **Pressure** = Second Order, **Density**, **Momentum**, **Energy**, k and ω = First Order Upwind (Second Order Upwind after the switch). **Monitors** → **Residuals**: all criteria 10^{-12}.

6.3 Summary and convergence criterion

Table 10: Solver settings for each speed.

	Supersonic, $M = 1.2$ and 1.47	Transonic, $M = 0.95$
Solver	density-based, implicit	pressure-based, Coupled, pseudo time
Main unknowns	density, momentum, energy, together	velocity and pressure, together; density from $p = \rho RT$
Faces	Roe-FDS flux	upwind for density, momentum, energy, k, ω; second-order pressure
Order	first order, then second order upwind	first order for 200 iterations, then second order
Gradients	least squares cell based	least squares cell based
Time step	CFL 1, then 5 (3 at the end of $M = 1.2$)	automatic pseudo time step
Viscous heating	always included	ticked by hand
Why	shock-dominated flow, information travels downstream	subsonic domain with a weak shock; the density-based solver swung

Convergence.

$$\Delta = \frac{\max C_D - \min C_D}{\text{mean } C_D} \text{ over the last 500 iterations} < 0.1\,\%. \tag{13}$$

Residuals do not show whether the drag has settled.

7 Step 5: Running the three cases

7.1 Steps for one Mach number

1. **Initialization** → **Hybrid** → **Initialize**. Hybrid initialisation builds a smooth first guess of the flow around the body from the boundary values, which is a much better start than a uniform field.
2. First order, CFL 1, for 100 iterations (**Run Calculation** → **Number of Iterations** → **Calculate**, or `/solve/iterate 100` in the console). Watch `cd-nose`: it falls from about 1.3–1.7 (the initial field) as the shock forms.
3. First order, CFL 5, to iteration 400 (density-based). The pressure-based run stays in first order for 200 iterations, with its automatic pseudo-time step.
4. Switch every equation to second order upwind and iterate until the criterion of Section 6 holds over the last 500 iterations. If the residuals start to grow, lower the CFL (5 → 3).
5. Save the case and the data (**File** → **Write** → **Case & Data**) with the Mach number in the name, `studyA_M1.47.cas.h5`; save the `cd-nose` history file under the same name.
6. Read the results: C_D and its pressure/viscous split from **Reports** → **Forces** on `wall_nose`; the maximum y^+ on both walls from **Reports** → **Surface Integrals** or a contour of **Wall Yplus**; Mach contours for the flow field (Section 8).

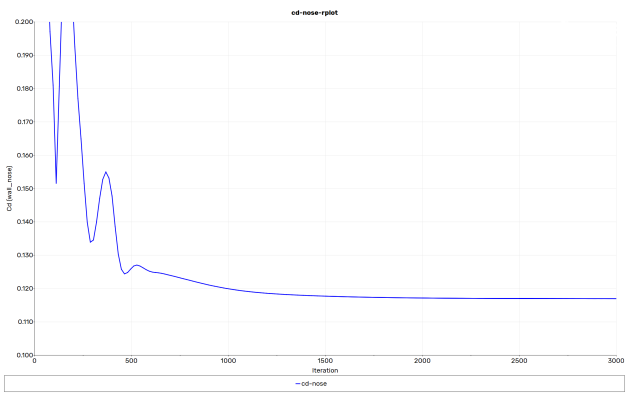
Next Mach number. Read the converged case, change p_∞, M and T_∞ on `inlet` and `farfield`, p_∞ and the backflow temperature on `outlet` (the turbulence values follow from I and μ_t/μ), redo **Reference Values** → **Compute from** `inlet`, and **initialise again** before step 2. Restarting $M = 1.2$ from the $M = 1.47$ solution went unstable after 180 iterations: the bow shock has to move too far. A fresh hybrid initialisation also resets the iteration count and the plots.

Check. Before calling a run converged: $\Delta < 0.1\,\%$ over the last 500 iterations, the maximum y^+ is below 1, and C_D recomputed from the force equals the reported value.

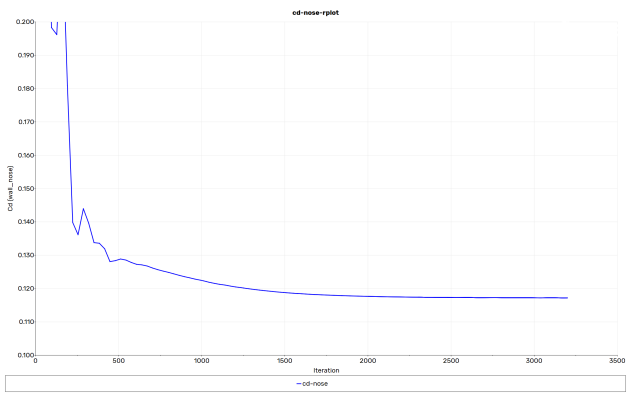
7.2 The runs of study A

Table 11: Run history. Δ is the C_D variation over the last 500 iterations.

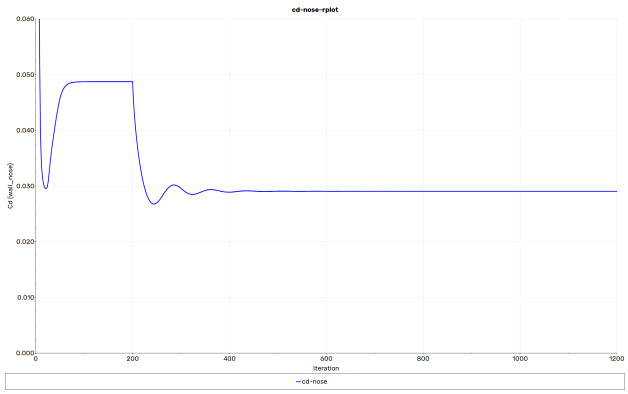
Run	Iterations	Order	Time step	Note
$M = 1.47$, density-based	1–100	first	CFL 1	start from hybrid initialisation
	101–400	first	CFL 5	
	401–3000	second	CFL 5	$\Delta = 0.032\,\%$ over 2501–3000
$M = 1.2$, density-based	1–100	first	CFL 1	fresh initialisation
	101–400	first	CFL 5	
	401–2400	second	CFL 5	continuity residual rose from $1.3 \cdot 10^{-3}$ to $8 \cdot 10^{-3}$
	2401–3200	second	CFL 3	residuals flat; $\Delta = 0.083\,\%$ over 2701–3200
$M = 0.95$, pressure-based	1–200	first	pseudo-time	C_D steady at 0.0488
	201–1200	second	pseudo-time	$\Delta = 0.009\,\%$ over 701–1200
$M = 0.95$, density-based	1–4300	first, then second	CFL 1–3	not converged, not used



(a) $M = 1.47$, density-based.



(b) $M = 1.2$, density-based.



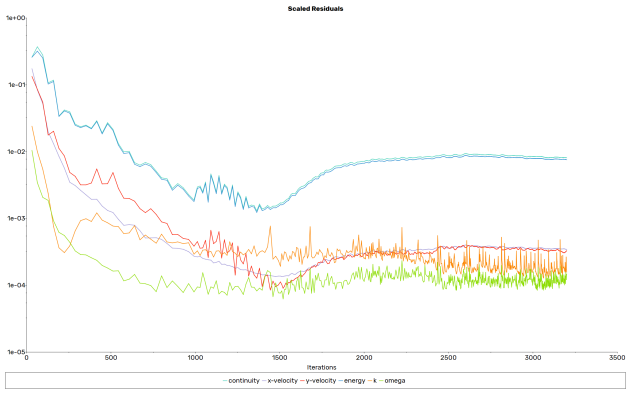
(c) $M = 0.95$, pressure-based.

Figure 6: The `cd-nose` monitor of the three converged runs (Fluent report plots). The vertical axis is fixed (0.10–0.20 above $M = 1$, 0–0.06 at $M = 0.95$), so the first iterations go off the top. Second order starts at iteration 400 in (a) and (b) and at iteration 200 in (c).

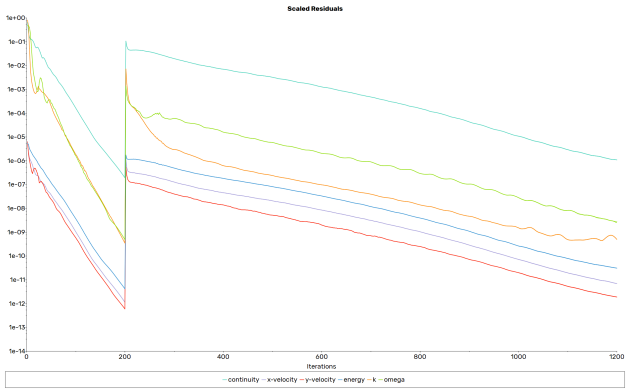
Each curve shows a first-order transient, a drop at the switch to second order (first order adds numerical dissipation, seen as extra drag), then a slow decay; the last per cent takes 1500–2000 iterations. First-order values, such as 0.0488 at $M = 0.95$, are only starting fields.



(a) $M = 1.47$, density-based.



(b) $M = 1.2$, density-based.



(c) $M = 0.95$, pressure-based.

Figure 7: Scaled residuals from Fluent. At $M = 1.47$ they level off at 10^{-4}–$3 \cdot 10^{-3}$ while C_D is already constant (Figure 6a), which is why convergence is judged on C_D.

8 Results of study A

8.1 Drag of the nose

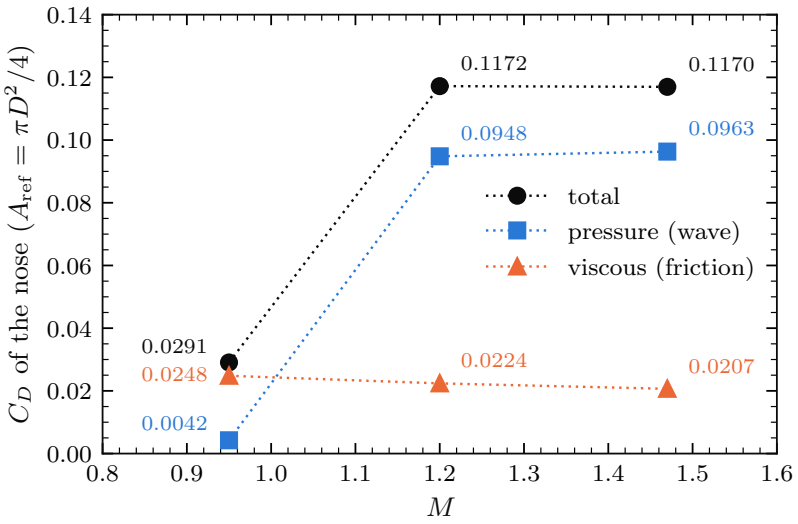
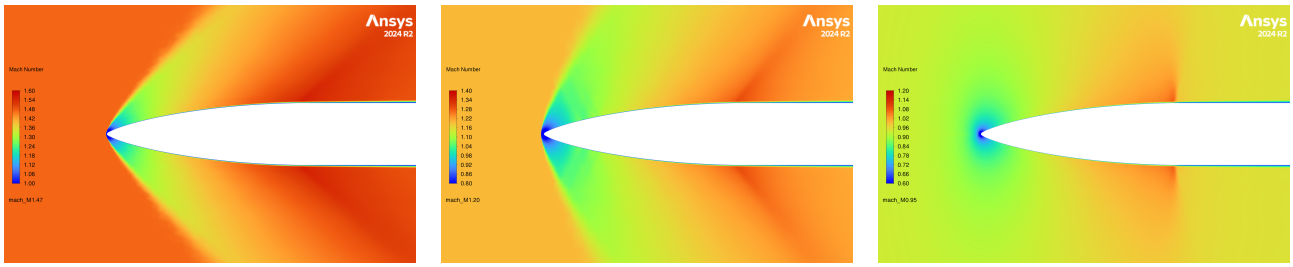


Figure 8: Drag coefficient of the nose (`wall_nose`) against Mach: pressure (wave), viscous and total. Drawn from the Fluent force reports of the three runs (Fluent cannot plot across separate runs); the dotted lines only join the three runs.

- **Supersonic.** The pressure part is 0.095–0.096, 81–82 % of the total, and nearly constant between $M = 1.2$ and 1.47.
- **Transonic.** The pressure part is only 0.0042: the overpressure at the tip is almost cancelled by the suction on the rear of the nose. Friction is 85 % of the total.
- **Friction** falls from 0.0248 to 0.0207 as Re_x rises and the wall gets hotter.

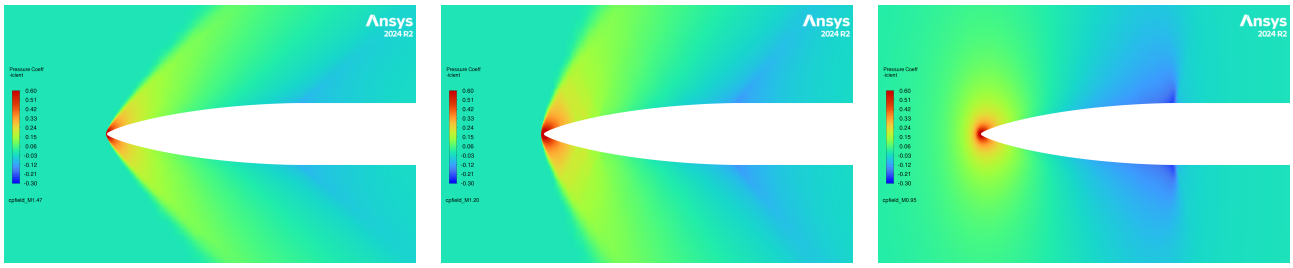
8.2 Flow field



(a) $M = 1.47$ (1.0–1.6). (b) $M = 1.2$ (0.8–1.4). (c) $M = 0.95$ (0.6–1.2).

Figure 9: Mach number contours from Fluent around the nose, mirrored about the axis; each run has its own colour scale.

- $M = 1.47$**.** The bow shock stands off the blunt tip, with subsonic flow behind it near the axis (dark blue). It weakens outwards to $M \approx 1.3$–1.4. The flow re-expands along the nose to $M \approx 1.55$ at the shoulder.
- $M = 1.2$**.** Same picture; the shock stands further off and is more curved. $M \approx 1.3$ at the shoulder.
- $M = 0.95$**.** No bow shock. The flow stagnates at the tip and accelerates along the nose; a supersonic pocket ($M \approx 1.05$–1.1) forms at the shoulder and ends in a weak shock.



(a) $M = 1.47$. (b) $M = 1.2$. (c) $M = 0.95$.

Figure 10: Pressure coefficient $C_p = (p - p_\infty)/q_\infty$ around the nose, from Fluent, on the same colour scale (-0.3 to 0.6) for all three runs.

Above $M = 1$ the high pressure behind the bow shock (red to yellow) covers the front half of the nose: that is the wave drag. At $M = 0.95$ the high pressure stays close to the tip and the rear of the nose is in suction (blue), so the two nearly cancel (Figure 10).

8.3 Checks

Table 12: Checks of the three runs.

	$M = 0.95$	$M = 1.2$	$M = 1.47$
Maximum y^+ on `wall_nose` / `wall_body`	0.51 / 0.28	0.60 / 0.30	0.71 / 0.29
Drag force F_x on `wall_nose`	42.32 N	254.75 N	349.42 N
`cd-nose` reported by Fluent	0.0291	0.1172	0.1170
Δ over the last 500 iterations	0.009 %	0.083 %	0.032 %

- $y^+ < 1$ on both walls at all three Mach numbers.
- Not done yet: a mesh-independence study (Section 10).

8.4 Pressure and friction along the nose

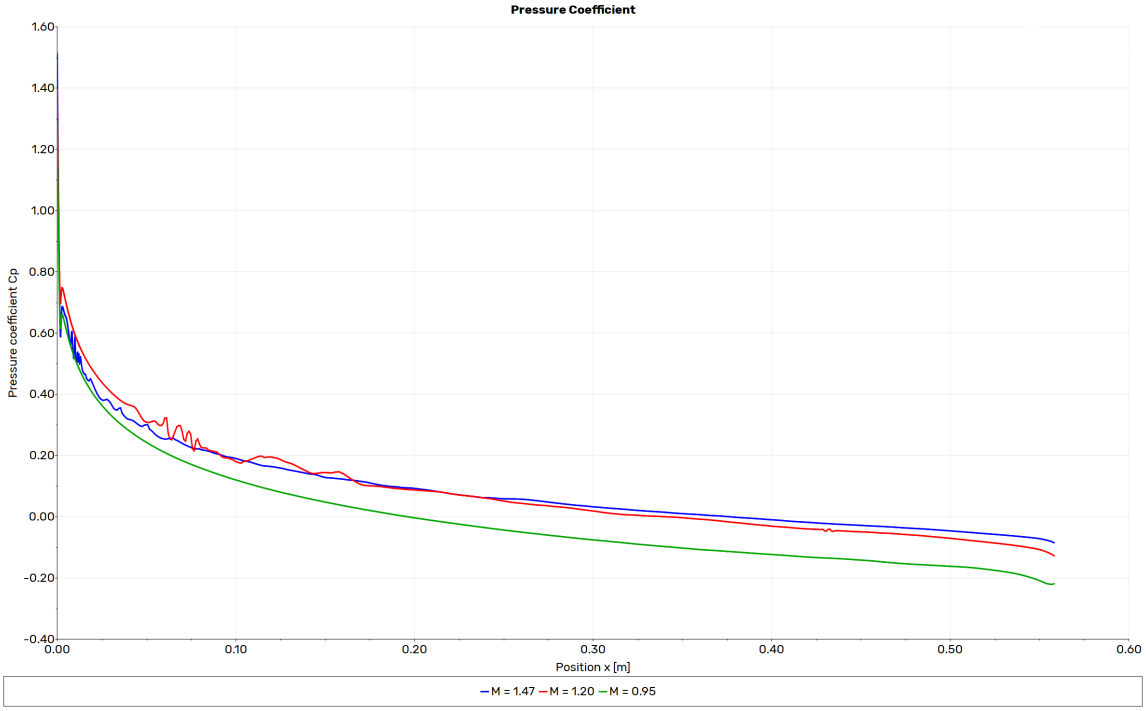


Figure 11: Pressure coefficient along `wall_nose` at the three Mach numbers (Fluent XY plot).

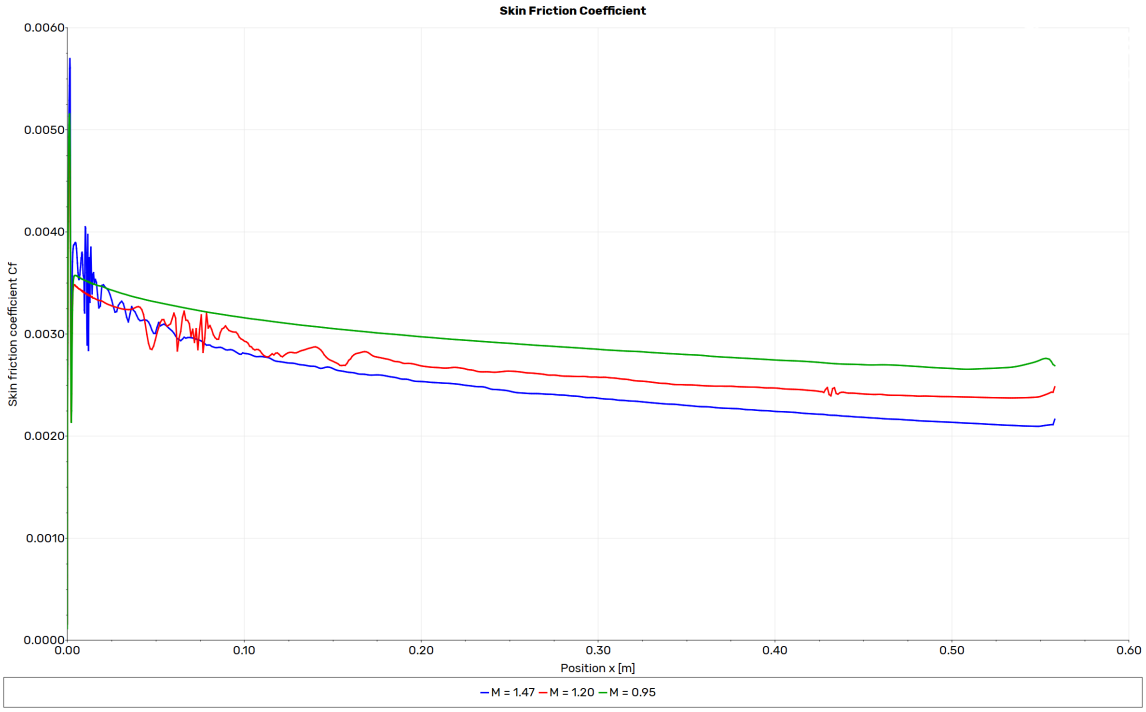


Figure 12: Skin-friction coefficient along `wall_nose` at the three Mach numbers (Fluent XY plot).

Pressure (Figure 11). It peaks at the tip, $C_p = 1.51$ ($M = 1.47$) and 1.40 ($M = 1.2$), within 0.4 % of the normal-shock value: the bow shock is captured correctly. Above $M = 1$ the wall stays above p_∞ up to $x \approx 340$–375 mm: that is the wave drag. At $M = 0.95$ it drops below p_∞ at $x \approx 195$ mm and

reaches $C_p = -0.22$ at the shoulder, so the suction cancels most of the push on the front. The wiggles at $x = 50$–170 mm ($M = 1.2$) are numerical and average out.

Friction (Figure 12). Highest just behind the tip ($\approx 5 \cdot 10^{-3}$), it falls to 2.2–$2.7 \cdot 10^{-3}$ at the shoulder as the boundary layer grows, and is lower at higher M.

9 Comparison with the Python results of report 2

Report 2 used two nose terms from the Python tool. They are compared here with Fluent at the same free stream and reference area.

- **Wave drag**: Euler simulations (no viscosity, pointed tip) interpolated to $L/D = 3.117$. Below $M = 1.1$ there were no Euler runs, so the tool uses an assumed ramp:

$$C_{D,w}(M) = C_{D,w}(1.1)\; s^2(3 - 2s), \qquad s = \frac{M - 0.8}{0.2} \text{ clipped to } [0, 1]. \tag{14}$$

  Compared with Fluent's pressure part.

- **Friction**: turbulent flat plate on the nose alone, smooth wall:

$$C_f = \frac{0.455}{(\log_{10} \mathrm{Re}_L)^{2.58}} \times \begin{cases} 1 - 0.1M^2, & M < 1, \\ (1 + 0.15M^2)^{-0.58}, & M > 1, \end{cases} \qquad C_{D,f} = C_f \frac{S_{\text{wet}}}{A_{\text{ref}}}, \quad S_{\text{wet}} = 0.2243\,\text{m}^2. \tag{15}$$

  Compared with Fluent's viscous part.

Table 13: Nose drag of study A: Python (report 2) and Fluent. Difference = Fluent / Python − 1.

	pressure (wave) C_D			friction (viscous) C_D		
M	Python	Fluent	difference	Python	Fluent	difference
0.95	0.0768	0.0042	−94.5 %	0.0236	0.0248	+5.2 %
1.2	0.0926	0.0948	+2.4 %	0.0225	0.0224	−0.5 %
1.47	0.0941	0.0963	+2.4 %	0.0210	0.0207	−1.4 %

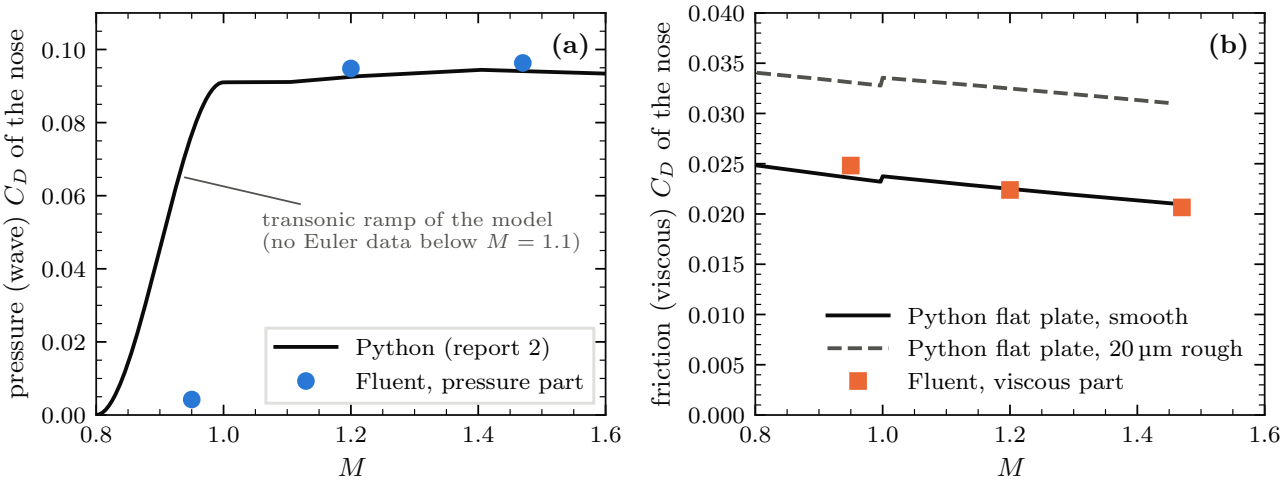


Figure 13: (a) Wave drag of the Python model against Mach, and Fluent's pressure part. (b) Python flat-plate friction of the nose along the boost of the report-2 trajectory (the Reynolds number changes with M), for a smooth wall and for the 20 µm roughness of the trajectory model, and Fluent's viscous part. The small step at $M = 1$ is where the Python formula switches from its subsonic to its supersonic compressibility correction.

Supersonic: $+2.4\,\%$ at both Mach numbers. Fluent has the real 4.5 mm rounded tip and a boundary layer that makes the nose act slightly fatter; both add a little pressure drag. With the sensitivity of report 2 ($\approx$ 7 m of apogee per per cent of wave drag), $+2.4\,\%$ costs about 17 m.

Friction: within $-1.4\,\%$ to $+5.2\,\%$. The flat plate models the nose friction well. The 20 µm roughness of the trajectory model adds 40–47 % (dashed line), which these smooth-wall runs do not check.

Transonic: Python is conservative. At $M = 0.95$ the ramp gives 0.0768 ($s = 0.75$, so 84 % of the $M = 1.1$ value); Fluent gives 0.0042. The drag rise comes later and steeper than assumed, so the drag of report 2 is too high between $M \approx 0.8$ and 1.0 and its apogee is on the safe side. Placing the rise needs a Fluent run at $M = 1.0$–1.1 (Section 10).

10 Conclusions and next runs

- Converged drag: $C_D = 0.0291$ / 0.1172 / 0.1170 at $M = 0.95$ / 1.2 / 1.47, with $y^+ < 1$ and $\Delta < 0.1\,\%$ in every run.
- The mesh is within the Ansys limits: skewness 0.92, orthogonal quality 0.237 (Meshing) / 0.077 (Fluent).
- Supersonic runs: density-based solver. Transonic run: pressure-based Coupled with Viscous Heating.
- Against Python: wave drag $+2.4\,\%$, friction within 5 %; the transonic ramp is conservative below $M = 1$.

Next runs, in order of priority.

1. **Mesh independence** at $M = 1.47$: coarse and fine mesh ($\times 1.5$ per direction each step), plus 1–2 pressure-gradient adaptions at the shock.
2. **Transonic rise**: $M = 1.05$ (pressure-based) and $M = 1.1$.
3. **Solver cross-check**: $M = 1.2$ with the pressure-based solver.
4. **Backflow temperature**: one run with T_0 instead of T_∞.
5. **Free-stream turbulence**: one run at $I = 0.1\,\%$, $\mu_t/\mu = 1$.
6. **Study B** (Von Kármán 630 + 20 mm): same steps; pull the coupler end from $x = 645.42$ mm to 2447.65 mm.
7. **Incidence**: 3D half model at 5° for $C_{N\alpha}$ and the centre of pressure ($y^+ \approx 30$–50 is enough there).

A Files

In the repository (`weisshorn-nose-cone`, paths from its root):

File	Contents
`CFD/geometry_650/nose_lvhaack_L561_c89_rn4.5mm.step, .csv`	Study-A design solid and profile points
`CFD/geometry_650/nose_vonkarman_L630_c20_rn4.5mm.step, .csv`	Study-B design solid and profile points
`CFD/geometry_650/studyA_domain2D_2024R2.dsco`	Study-A 2D flow domain (Discovery 2024 R2)
`CFD/studyA/README.md, studyA_log.html`	Run log of study A
`CFD/studyA/freestream_bc.py`	Free stream, boundary values, first-cell height and C_D from the forces
`CFD/studyA/cd_history/*.csv`	C_D on `wall_nose` per iteration, all four runs
`CFD/studyA/plots/*.png`	Fluent pictures: C_D, residuals, Mach contours
`CFD/studyA/report_figures.py`	Figures of this report and the comparison with report 2
`CFD/mcp/`	Claude Code + Fluent MCP set-up for the DLL remote PCs (Appendix B)
`CFD/source/`	Source of this report

B Driving Fluent from Claude Code on the DLL remote PCs

The study-A runs were driven by Claude Code through an MCP server connected to the running Fluent session. This is optional: every step of this report can be done in the Fluent GUI. The set-up is in `CFD/mcp/` and follows 2026_C_FD_REPORT-2_MCP-FLUENT, with two differences: the MCP server is `CFD/mcp/server.py` of this repository (the guide does not give the URL of its own), and the Claude Code home is moved to the network drive with the `CLAUDE_CONFIG_DIR` variable, because the guide's junction cannot point to a network drive.

Set-up, every session.

1. Put the repository in the Documents folder of the network home, which survives the reset of `C:\` (clone it from there, or download the branch as a ZIP from GitHub).
2. In a normal PowerShell window run

   ```
   powershell -ExecutionPolicy Bypass -File <repo>\CFD\mcp\bootstrap-vm.ps1
   ```

   Each step is skipped when its result is already there: the first run takes about 10 minutes, later runs seconds. It finds the newest Fluent installation and Python, installs PyFluent 0.43 and `mcp` 2.x from the wheel cache on the network drive, installs or restores Claude Code, writes `.mcp.json` (without a BOM) and starts Claude Code in the repository.
3. Start Fluent yourself (Workbench or the Fluent launcher, working directory `CFD/runs`). In its console type `define/beta-feature-access` (answer yes), then `server/start-server`. Fluent 2024 R2 refuses `start-server` until the beta features are on. Fluent writes a `server_info*.txt` file.
4. Ask Claude Code to connect to that file (`connect_to_fluent`). The `launch_fluent` tool can also start a new session, but it always picks the newest installed version (2026 R1 on these PCs), not 2024 R2.

Restarting.

- **Claude Code only:** type `/exit`, then run the setup script again, or start it by hand with `claude -continue` from the repository folder. `-continue` reopens the last conversation; inside Claude Code, `/resume` picks an older one. The history is kept on the network drive, so it survives the restart. Fluent can stay open.
- **MCP connection only:** type `/mcp`, choose `fluent` and **Reconnect**. Then ask Claude to run `connect_to_fluent` again, with the `CFD/runs` folder. PyFluent takes up to a minute to load.
- **Fluent restarted:** type `server/start-server` again in the new Fluent console, then reconnect. Each Fluent session writes its own `server_info` file, and Claude uses the newest one in the folder.

Tools. The server offers 13 tools. Session: `launch_fluent`, `connect_to_fluent`, `exit_fluent`. Files: `read_case`, `read_case_and_data`, `write_data`. Solver: `initialize`, `iterate`, `execute_tui`, `scheme_eval`. Reports: `get_case_summary`, `get_residuals_summary`, `screenshot`. The tool time-out is set to one hour so that long `iterate` calls finish.

Things to know.

- The server drives the solver only; geometry and meshing stay manual.
- Console commands with keyword menus are easy to get wrong. Let Claude Code run iterations and read results, but enter boundary values in the panels and check them with `/file/write-bc`.
- One `iterate` call hung with Fluent waiting at its “number of iterations” prompt; `/solve/iterate n` through `execute_tui` works.
- The copy of Claude Code on the PC is a separate installation: it starts without the context of the sessions on a laptop, so give it the case description (this report) at the start.